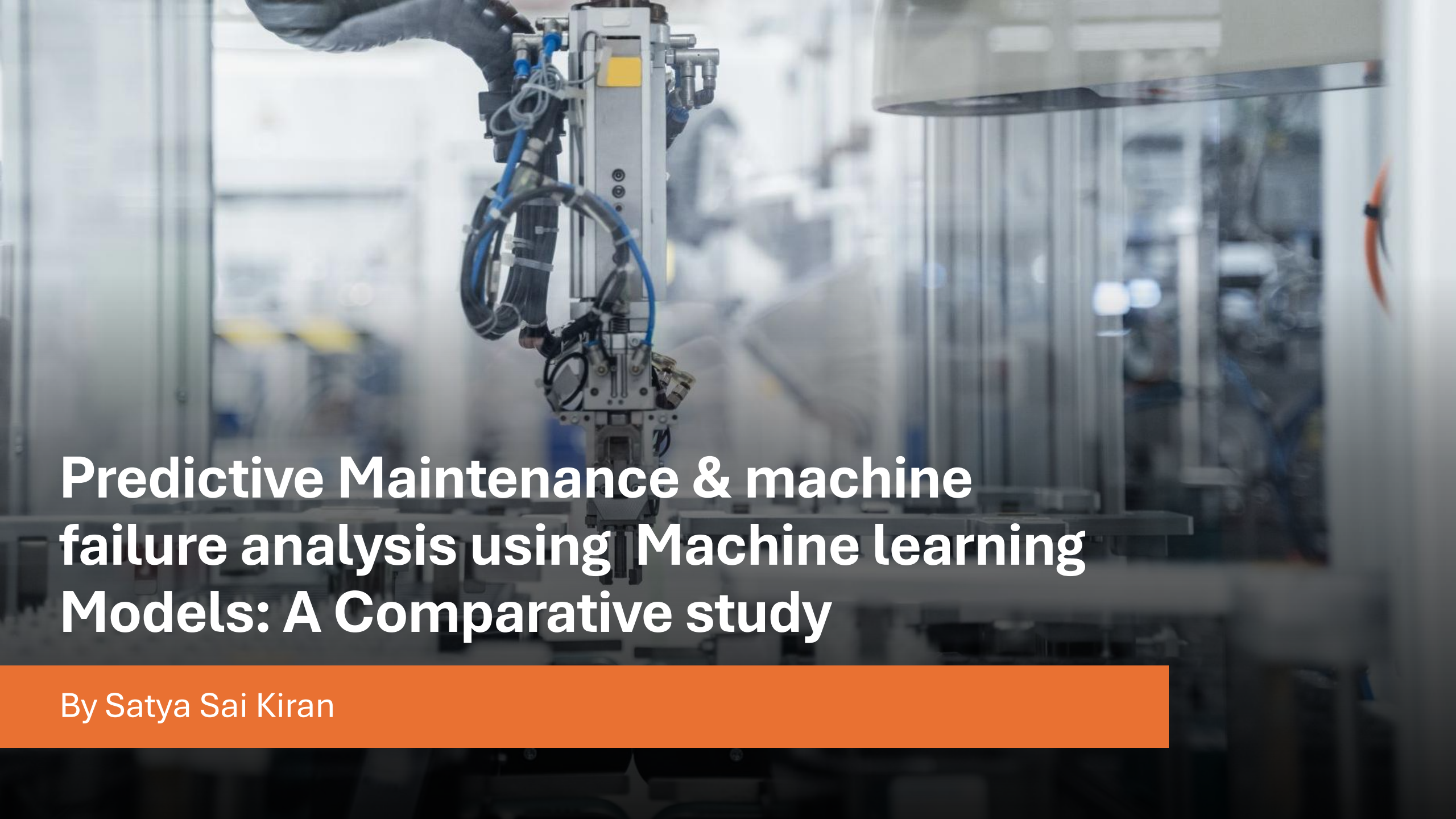


Predictive Maintenance & machine failure analysis using Machine learning Models: A Comparative study

By Satya Sai Kiran

Data Set

- Dataset is sourced from UC Irvine Machine Learning Repository under the name **AI4I 2020 Predictive Maintenance Dataset.**
- https://archive.ics.uci.edu/dataset/601/ai4i+2020+predictive+maintenance+dataset

UC Irvine Machine Learning Repository



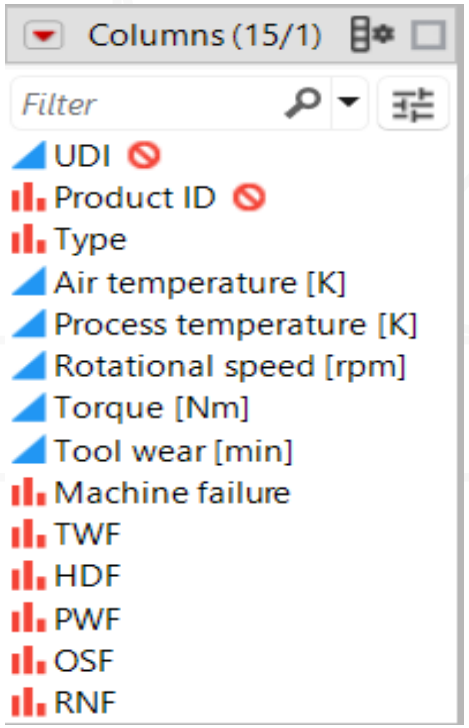
- It is a synthetic dataset that reflects real predictive maintenance data encountered in industry.
- The dataset consists of 10 000 data points stored as rows with 14 features in columns.
- Features include Product ID, Air and process temperatures, RPM, torque, tool wear, machine failure and five failure modes Tool wear(TWF), Power(PWF), Heat dissipation(HDF), Overstrain(OSF), Random(RNF) etc.,



Microsoft Excel
ma Separated Vali

Software package & Dataset preparation

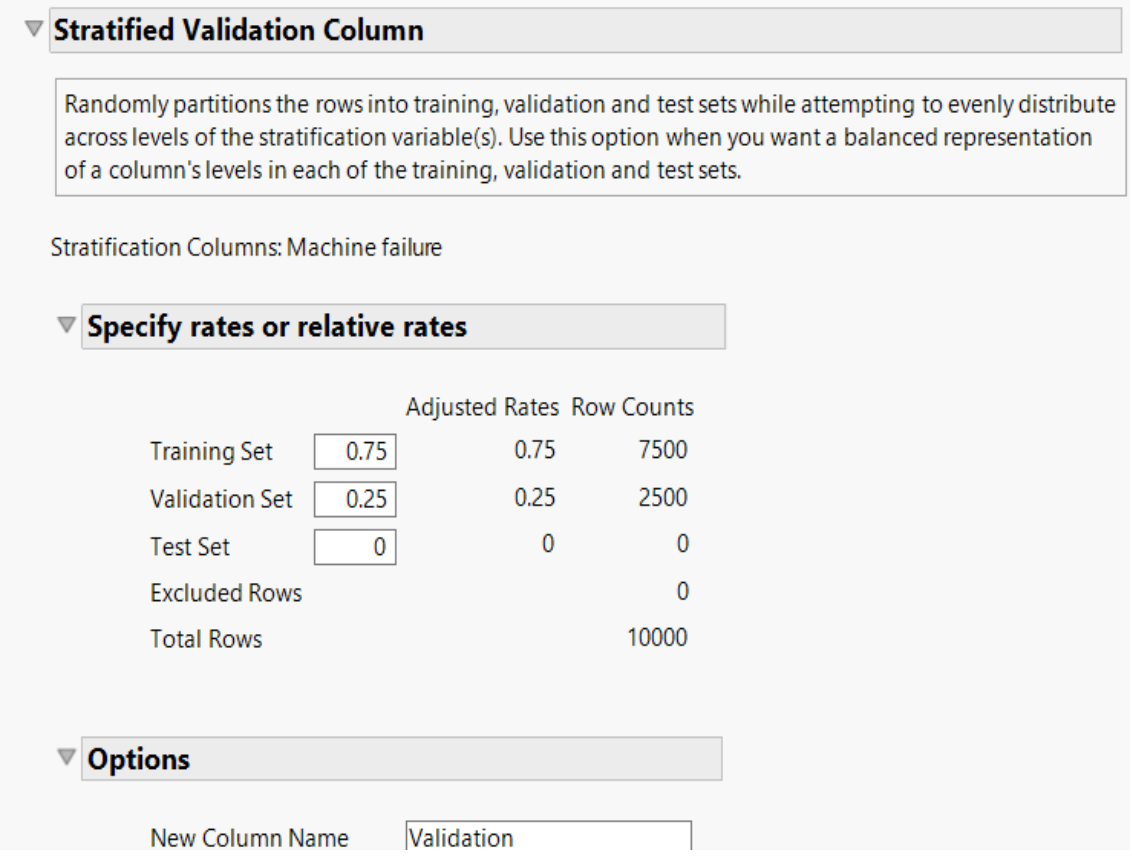
We have used JMP statistical software student version 19 to create the machine learning models and run the results



- Nominal: Product ID, Type, Machine failure, TWF, HDF, PWF, OSF, RNF.
- Continuous: Air temperature [K], Process temperature [K], Rotational speed [rpm], Torque [Nm], Tool wear [min].
- UDI and Product ID columns are **Excluded** These should not be used as predictors because they contain unique values that lead to overfitting.

Data set pre-processing

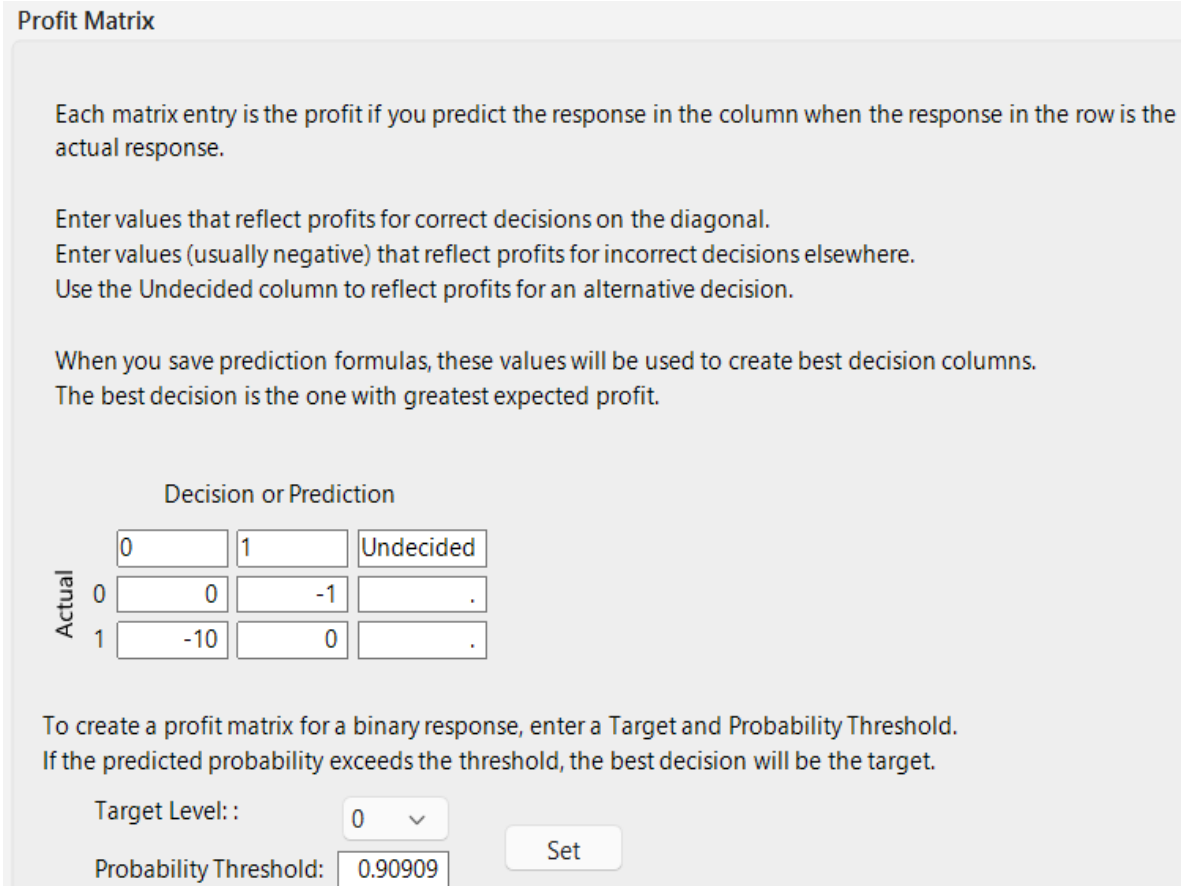
- A **validation column** is inserted from predictive modelling tab to separate the data into training and validation data sets, **75% and 25%** respectively.
- **Stratified Validation** Approach is chosen and Machine failure is selected as the stratification column- This ensures that failures are distributed evenly (in the same 3.4% ratio) between training set and testing set.



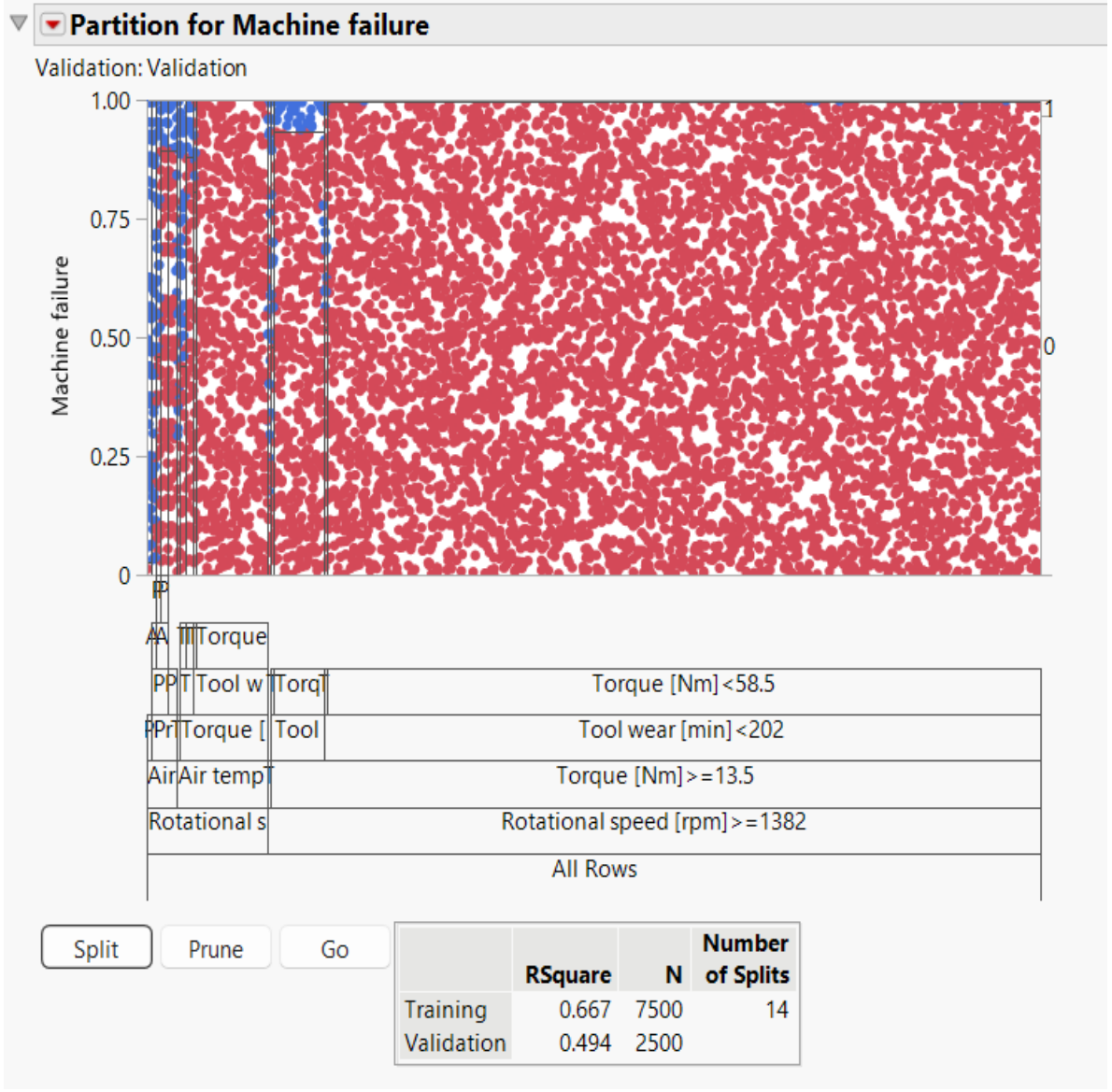
	UDI	Product ID	Type	Air temperature [K]	Process temperature [K]	Rotational speed [rpm]	Torque [Nm]	Tool wear [min]	Machine failure	TWF	HDF	PWF	OSF	RNF	Validation
7485	9970	L57149	L	298.4	308.1	1491	50	156	0	0	0	0	0	0	Training
7486	9971	H39384	H	298.4	308.1	1891	24.7	158	0	0	0	0	0	0	Training
7487	9974	L57153	L	298.5	308.2	1444	40.5	170	0	0	0	0	0	0	Training
7488	9975	L57154	L	298.6	308.2	1361	68.2	172	1	0	0	1	1	0	Training
7489	9976	M24835	M	298.6	308.3	1589	34.1	174	0	0	0	0	0	0	Training
7490	9980	L57159	L	298.5	308.3	1334	49.8	186	0	0	0	0	0	0	Training
7491	9981	L57160	L	298.6	308.4	1559	36.6	188	0	0	0	0	0	0	Training
7492	9982	L57161	L	298.7	308.5	1353	47	190	0	0	0	0	0	0	Training
7493	9983	M24842	M	298.7	308.4	1507	45.7	192	0	0	0	0	0	0	Training
7494	9985	H39398	H	298.7	308.4	1432	42.8	197	0	0	0	0	0	0	Training
7495	9987	M24846	M	298.8	308.6	1507	42.3	207	0	0	0	0	0	0	Training
7496	9988	M24847	M	298.8	308.6	1500	37.1	210	0	0	0	0	0	0	Training
7497	9992	M24851	M	298.9	308.4	1827	26.1	5	0	0	0	0	0	0	Training
7498	9994	L57173	L	298.8	308.4	1401	47.3	10	0	0	0	0	0	0	Training
7499	9996	M24855	M	298.8	308.4	1604	29.5	14	0	0	0	0	0	0	Training
7500	9997	H39410	H	298.9	308.4	1632	31.8	17	0	0	0	0	0	0	Training
7501	5	L47184	L	298.2	308.7	1408	40	9	0	0	0	0	0	0	Validation
7502	13	M14872	M	298.6	309.1	1339	51.1	34	0	0	0	0	0	0	Validation
7503	29	L47208	L	299.1	309.4	1439	44.2	82	0	0	0	0	0	0	Validation
7504	49	H29462	H	298.8	309.2	1425	53.9	135	0	0	0	0	0	0	Validation
7505	65	L47244	L	299	309.1	1523	38.6	177	0	0	0	0	0	0	Validation
7506	74	L47253	L	299	309.1	1592	35	200	0	0	0	0	0	0	Validation
7507	113	M14972	M	298.8	308.7	1498	39.5	89	0	0	0	0	0	0	Validation
7508	136	M14995	M	298.8	308.5	1711	28.5	140	0	0	0	0	0	0	Validation
7509	144	H29557	H	298.6	308.4	1407	50.5	164	0	0	0	0	0	0	Validation
7510	161	L47340	L	298.4	308.2	1282	60.7	216	1	0	0	0	1	0	Validation
7511	168	H29581	H	298.4	308.3	1458	42.9	15	0	0	0	0	0	0	Validation
7512	304	M15163	M	297.8	308.4	1479	43.5	143	0	0	0	0	0	0	Validation
7513	314	L47493	L	298	308.6	1313	50.8	170	0	0	0	0	0	0	Validation
7514	322	L47501	L	297.8	308.4	1518	32.5	190	0	0	0	0	0	0	Validation
7515	341	L47520	L	297.4	308.1	1474	39.4	20	0	0	0	0	0	0	Validation
7516	346	L47525	L	297.5	308.3	1585	32.9	32	0	0	0	0	0	0	Validation
7517	360	M15219	M	297.6	308.3	1567	34.6	71	0	0	0	0	0	0	Validation
7518	363	H29776	H	297.6	308.3	1640	30.9	79	0	0	0	0	0	0	Validation

Data set pre-processing

Defined Misclassification costs through profit matrix- we define to JMP that a **False Negative** (missing a failure) is much more expensive than a **False Positive** (a false alarm)



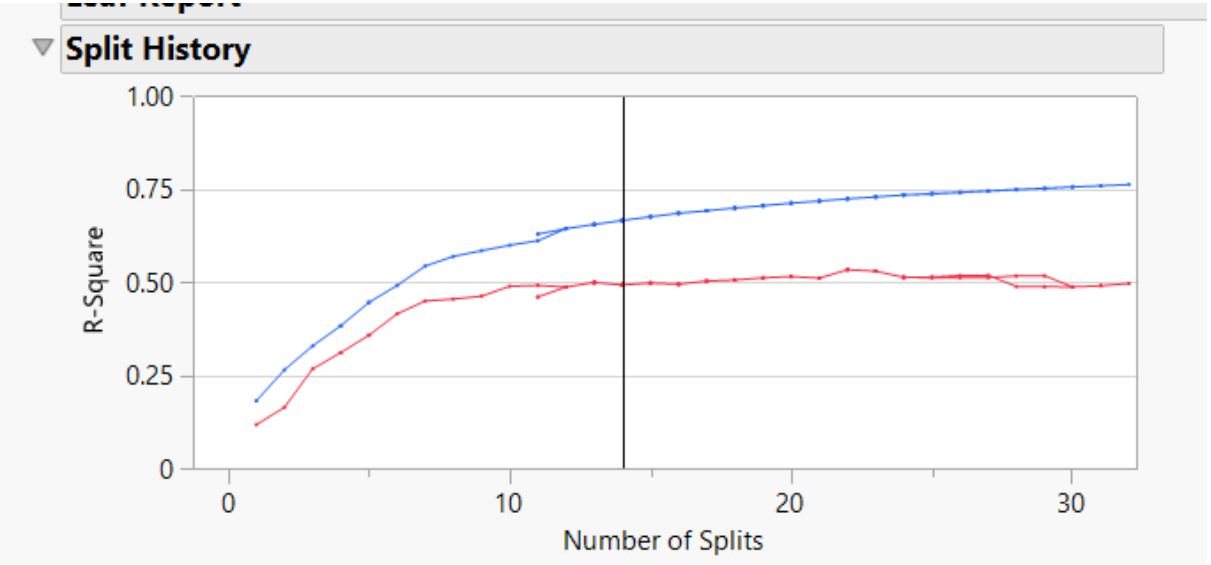
ML Model 1: Partition /Decision Tree



- **Y, Response**: Machine failure (binary).
- **X, Factors**: Type, Air temperature [K], Process temperature [K], Rotational speed [rpm], Torque [Nm], Tool wear [min].
- **Validation** : validation column

Splitting is done more than 20 times, and is pruned 6 times, as beyond 14 splits the validation performance R2- **0.494-0.5**, doesn't improve further

So, Optimum split is 14 times



Fit Details

Measure	Training	Validation	Definition
Entropy RSquare	0.6667	0.4940	1-Loglike(model)/Loglike(0)
Generalized RSquare	0.6989	0.5311	(1-(L(0)/L(model))^(2/n))/(1-L(0)^(2/n))
Mean -Log p	0.0493	0.0751	∑ -Log(ρ[j])/n
RASE	0.1157	0.1438	√ ∑(y[j]-ρ[j])²/n
Mean Abs Dev	0.0271	0.0355	∑ |y[j]-ρ[j]|/n
Misclassification Rate	0.0183	0.0284	∑ (ρ[j]≠ρMax)/n
N	7500	2500	n

Confusion Matrix

Training

Actual Machine failure	Predicted Count 0	1
0	7183	63
1	74	180

Validation

Actual Machine failure	Predicted Count 0	1
0	2381	34
1	37	48

Actual Machine failure	Predicted Rate 0	1
0	0.991	0.009
1	0.291	0.709

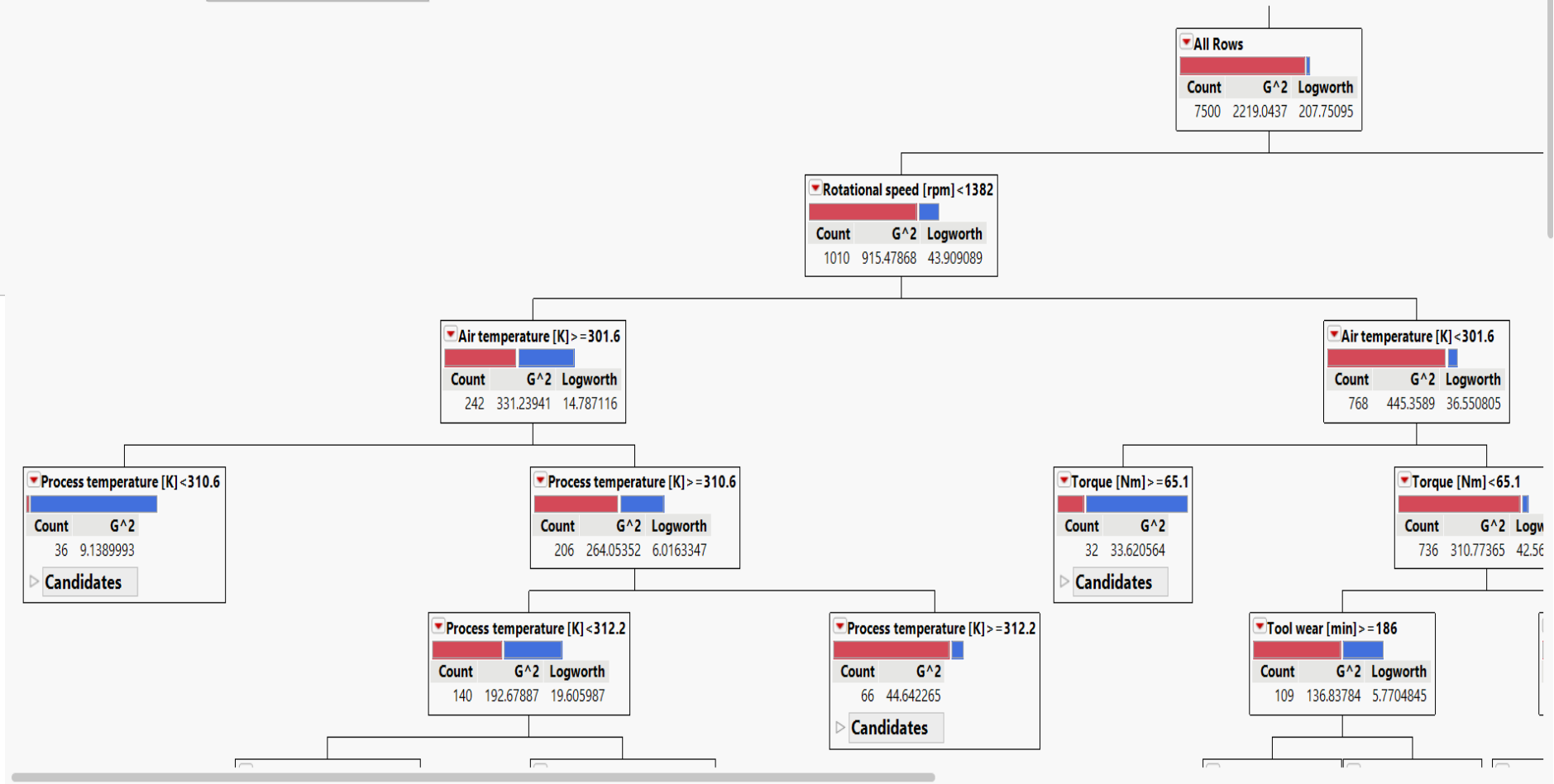
Actual Machine failure	Predicted Rate 0	1
0	0.986	0.014
1	0.435	0.565

Fit Details

Decision tree or partition

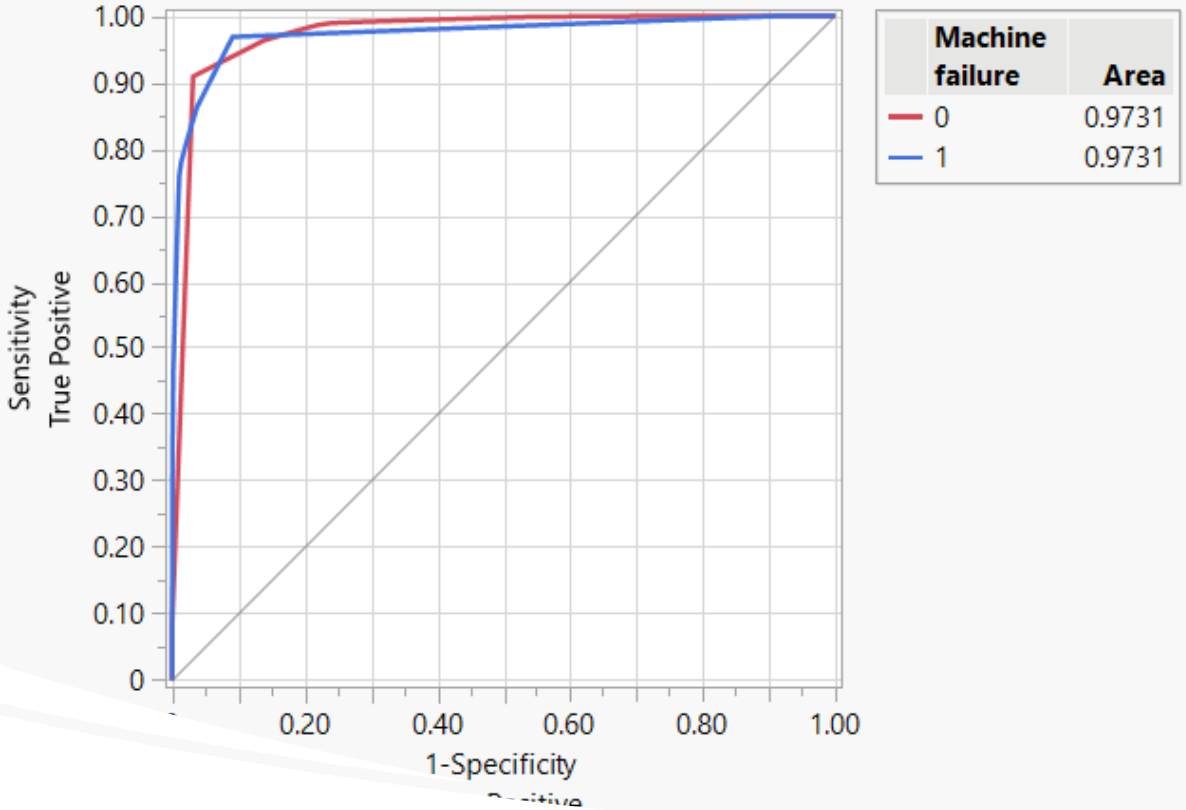
Split Prune Go

	RSquare	N	Number of Splits
Training	0.667	7500	14
Validation	0.494	2500	

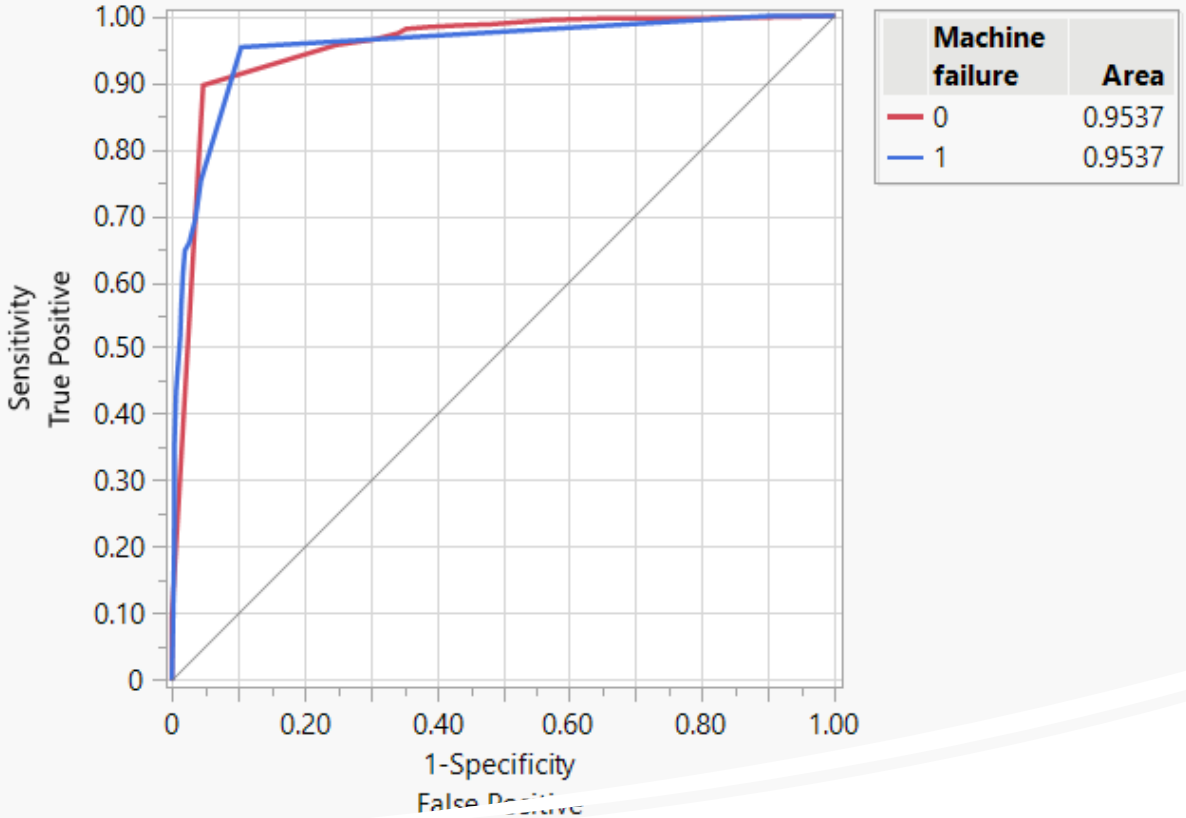


Decision Matrix

Receiver Operating Characteristic on Training Data



Receiver Operating Characteristic on Validation Data



ROC plot and Area under curve for partition

- **Good performance**: $R^2 \approx 0.48–0.67$ and AUC > 0.90 on validation shows the tree captures substantial signal.

Mild overfitting: Training R^2 (0.667) > Validation R^2 (0.484), but gap is reasonable for a tree with 14 splits.

Partition-Results Interpretation

- **First split on Torque ≥ 48.35 Nm**: High torque increases mechanical load
- **Tool wear ≥ 205 min**: Near the **TWF** threshold (200–240 min tool wear failure)
- **Rotational speed < 1382 rpm**: Matches **HDF** rule (heat dissipation failure: speed < 1380 rpm + small temp difference).
- **Process temp ≥ 309.2 K** and **Air temp ≥ 300.04 K**: Process temp should be ~10K hotter than air temp. When air temp is high (≥300.04) AND process temp is high (≥309.2), the **difference** is small → HDF condition.

1. RULE 1: Torque ≥ 48 Nm AND Tool wear ≥ 205 min → Failure risk ↑
2. RULE 2: Torque ≥ 48 Nm AND Rot speed < 1382 rpm → Failure risk ↑
3. RULE 3: Low Rot speed + High Air temp + High Process temp → Failure risk ↑ (HDF)

- **INSIGHT**: Engineers can use these exact thresholds to set alarms/inspections.

ML Model 2- Random Forest (Bootstrap Forest) – overall machine failure

- The model is very good at correctly predicting **non-failures** (class 0) with error rates <1% on both training and validation.
- For **failures** (class 1), it still catches the majority, but some failures are missed (recall ≈ 56–72% depending on train/validation

Bootstrap Forest for Machine failure

Specifications

Target	Machine failure	Training Rows:	7500
Validation Column:	Validation	Validation Rows:	2500
		Test Rows:	0
Number of Trees in the Forest:	300	Number of Terms:	6
Number of Terms Sampled per Split:	3	Bootstrap Samples:	7500
		Minimum Splits per Tree:	10
		Minimum Size Split:	10

Overall Statistics

Measure	Training	Validation	Definition
Entropy RSquare	0.7709	0.5898	1-Loglike(model)/Loglike(0)
Generalized RSquare	0.7963	0.6253	(1-(L(0)/L(model))^(2/n))/(1-L(0)^(2/n))
Mean -Log p	0.0339	0.0609	∑ -Log(ρ[j])/n
RASE	0.0957	0.1260	√ ∑(y[j]-ρ[j])²/n
Mean Abs Dev	0.0249	0.0348	∑ |y[j]-ρ[j]|/n
Misclassification Rate	0.0101	0.0184	∑ (ρ[j]≠ρMax)/n
N	7500	2500	n

Number of Trees
22

Confusion Matrix

Training

Actual Machine failure	Predicted Count 0	1
0	7240	6
1	70	184

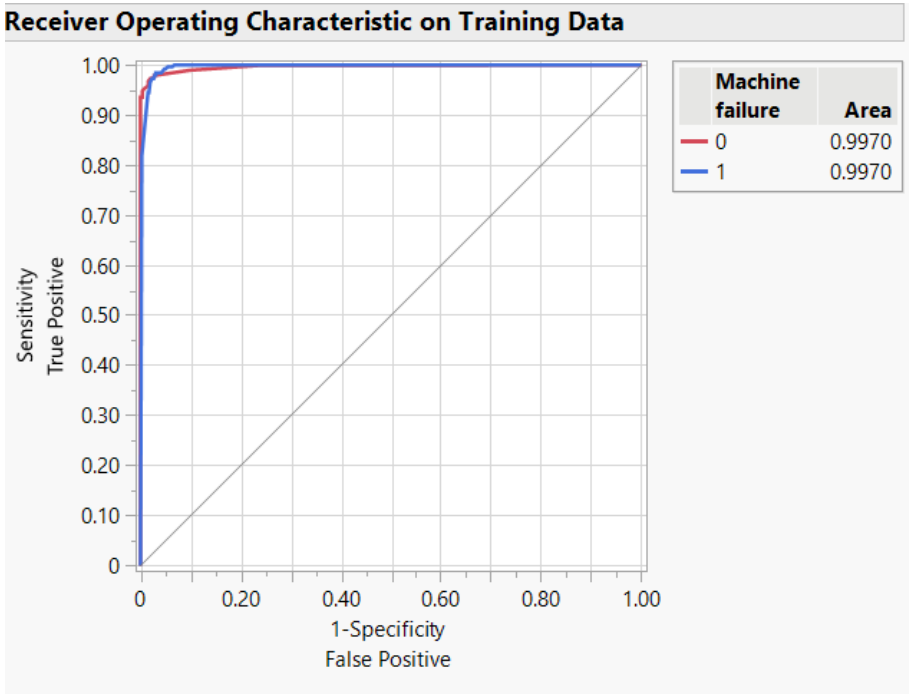
Validation

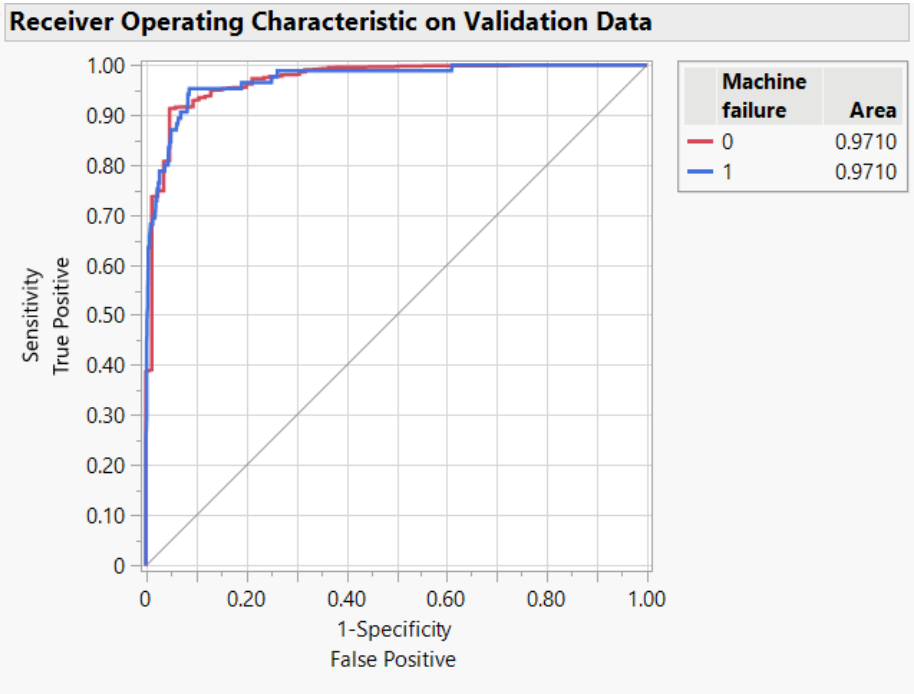
Actual Machine failure	Predicted Count 0	1
0	2406	9
1	37	48

Actual Machine failure	Predicted Rate 0	1
0	0.999	0.001
1	0.276	0.724

Actual Machine failure	Predicted Rate 0	1
0	0.996	0.004
1	0.435	0.565

INTERPRETATION OF FOREST MODEL RESULTS





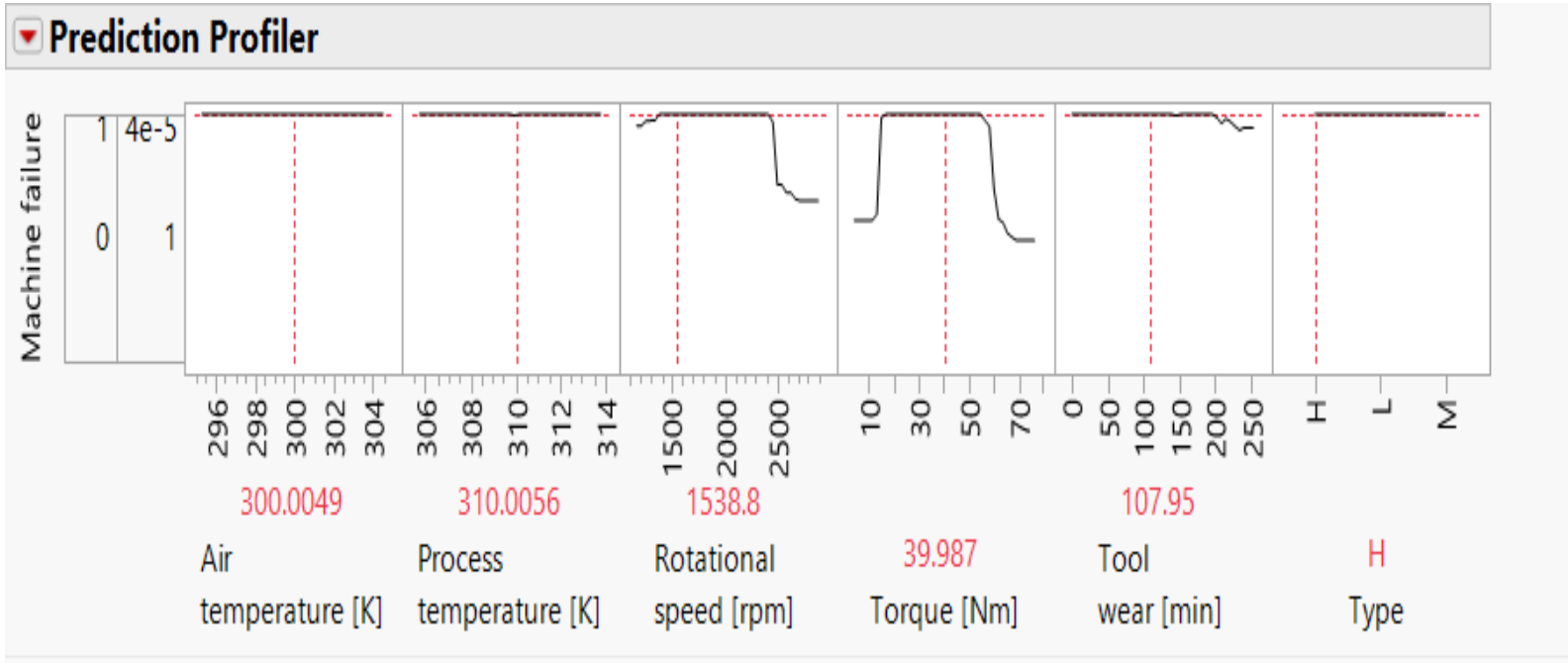
- AUC close to 1.0 on training indicates the forest can almost perfectly separate failures from non-failures.
- AUC ≈ 0.971 on validation is still excellent and confirms very strong discriminative power.

The **Bootstrap Forest is an excellent model** for predicting machine failure in this dataset:

High overall accuracy (~94–96% on validation).

Very high AUC (~0.97), meaning strong ranking ability.

Good but not perfect recall on failure class (around 56% on validation) with high precision (≈0.84).



The prediction profiler says that **torque**, **tool wear**, and **rotational speed** are the main drivers of the machine-failure probability, while the temperatures and product type have smaller, more specific effects.

Observations from Bootstrap Forest model

- "**high torque + high wear**" is the region where failures cluster. So, forest uses these two variables most heavily when splitting the data and predicting failure.
- Forest learns that "low speed together with small temp difference" corresponds to failure.
- Power failure (PWF) is linked to the product of torque and rotational speed (power). When power is too low or too high, failures are triggered. speed modifies both heat dissipation and power.

ML Model 3- Boosted tree for machine failure

- Sensitivity (Recall for failure) Model catches about **53% of actual failures.**
- Specificity- Excellent at identifying non-failures 99.7%

Boosted Tree for Machine failure Student Edition

Specifications

Target	Machine failure	Number of training rows:	7500
Validation Column:	Validation	Number of validation rows:	2500
Number of Layers:	42		
Splits per Tree:	16		
Learning Rate:	0.146		
Overfit Penalty:	0.0001		

Overall Statistics

Measure	Training	Validation	Definition
Entropy RSquare	0.8286	0.5779	1-Loglike(model)/Loglike(0)
Generalized RSquare	0.8489	0.6138	(1-(L(0)/L(model))^(2/n))/(1-L(0)^(2/n))
Mean -Log p	0.0254	0.0626	∑ -Log(ρ[j])/n
RASE	0.0738	0.1256	√ ∑(y[j]-ρ[j])²/n
Mean Abs Dev	0.0185	0.0309	∑ |y[j]-ρ[j]|/n
Misclassification Rate	0.0061	0.0192	∑ (ρ[j]≠ρMax)/n
N	7500	2500	n

Confusion Matrix

Training

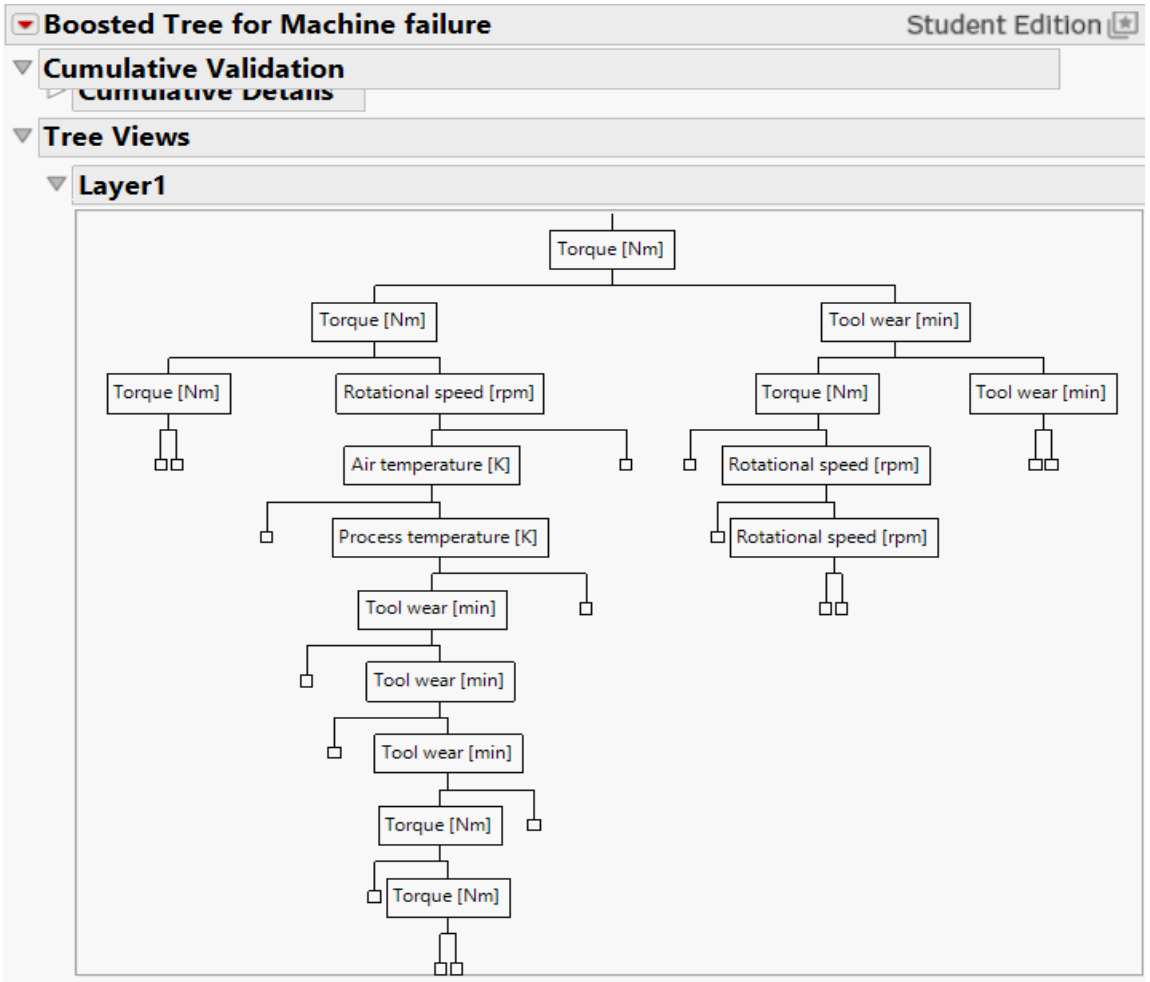
Actual Machine failure	Predicted Count 0	1
0	7245	1
1	45	209

Actual Machine failure	Predicted Rate 0	1
0	1.000	0.000
1	0.177	0.823

Validation

Actual Machine failure	Predicted Count 0	1
0	2407	8
1	40	45

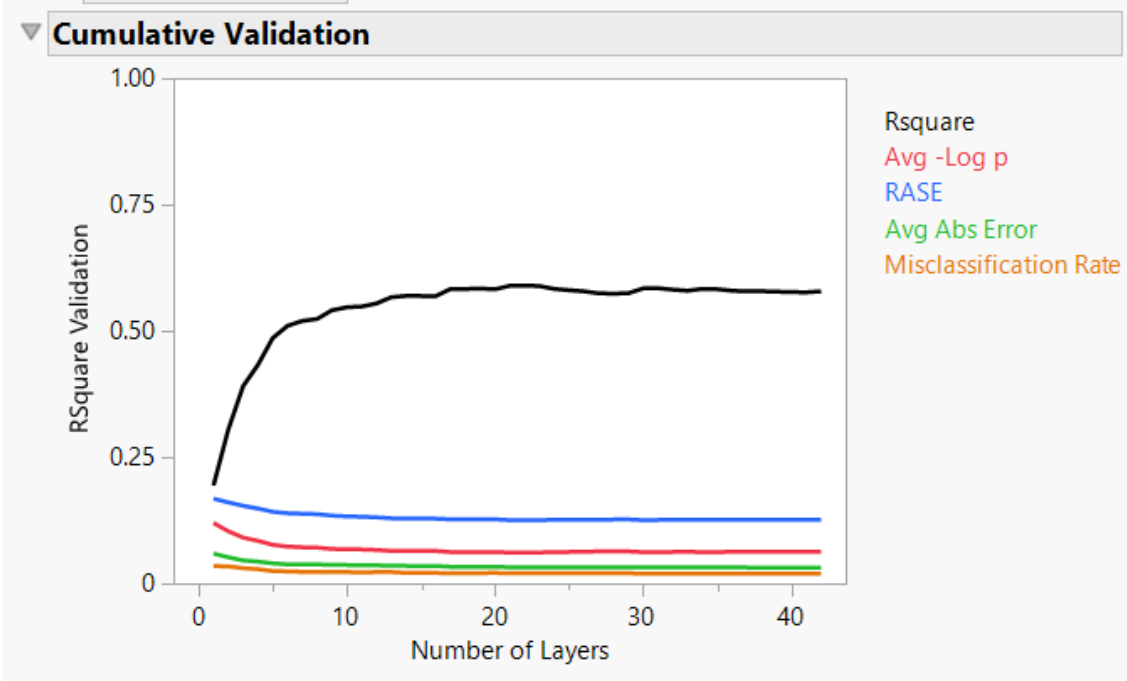
Actual Machine failure	Predicted Rate 0	1
0	0.997	0.003
1	0.471	0.529



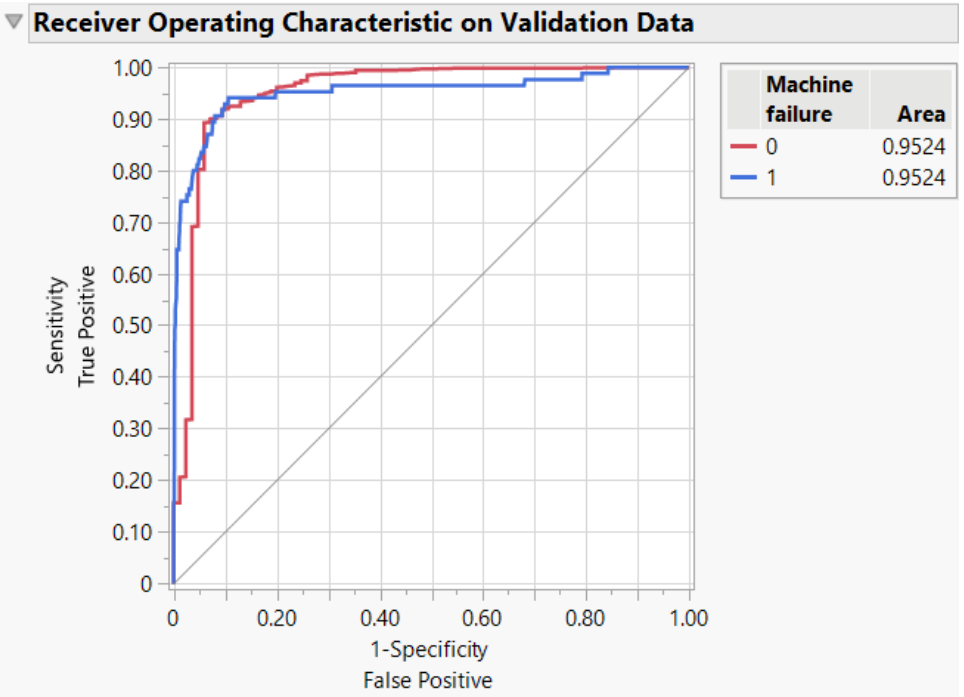
Interpretation of Boosted Tree Results

- Training $R^2 \approx$ **0.85**
- Validation $R^2 \approx$ **0.61**

This indicates: Model explains **~61% of variation** in failure probability on unseen data.







Validation AUC = **0.952**
This is **excellent discrimination power**

Comparative analysis of ML models

Boosted Tree wins on R^2 and misclassification rate. **Bootstrap Forest** excels at AUC. **Decision Tree** is competitive but simplest.

Metric	Decision Tree	Bootstrap Forest	Boosted Tree	Best Model
R^2	0.484	0.55	0.578	Boosted
Training AUC	0.973	0.997	0.981	Forest
Validation AUC	0.917	0.971	0.924	Forest
Misclassification Rate	6–7%	5.72%	3.9%	Boosted
Failure Precision	~80–85% (est.)	84.2%	92.3% (42/46, est.)	Boosted
Training Rows / Val Rows	7500 / 2500	7500 / 2500	7500 / 2500	—
Splits / Trees	14 splits	22 trees	16 learning rate	—

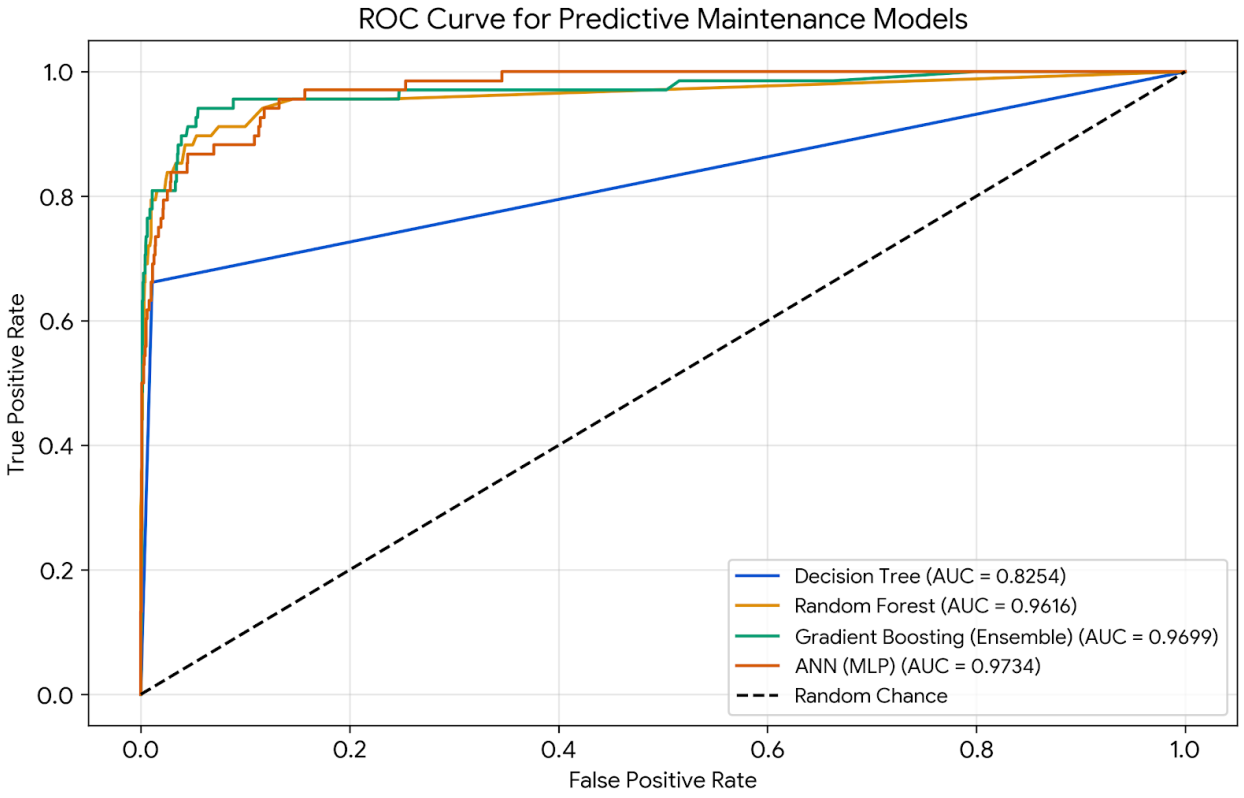
Model Performance Ranking

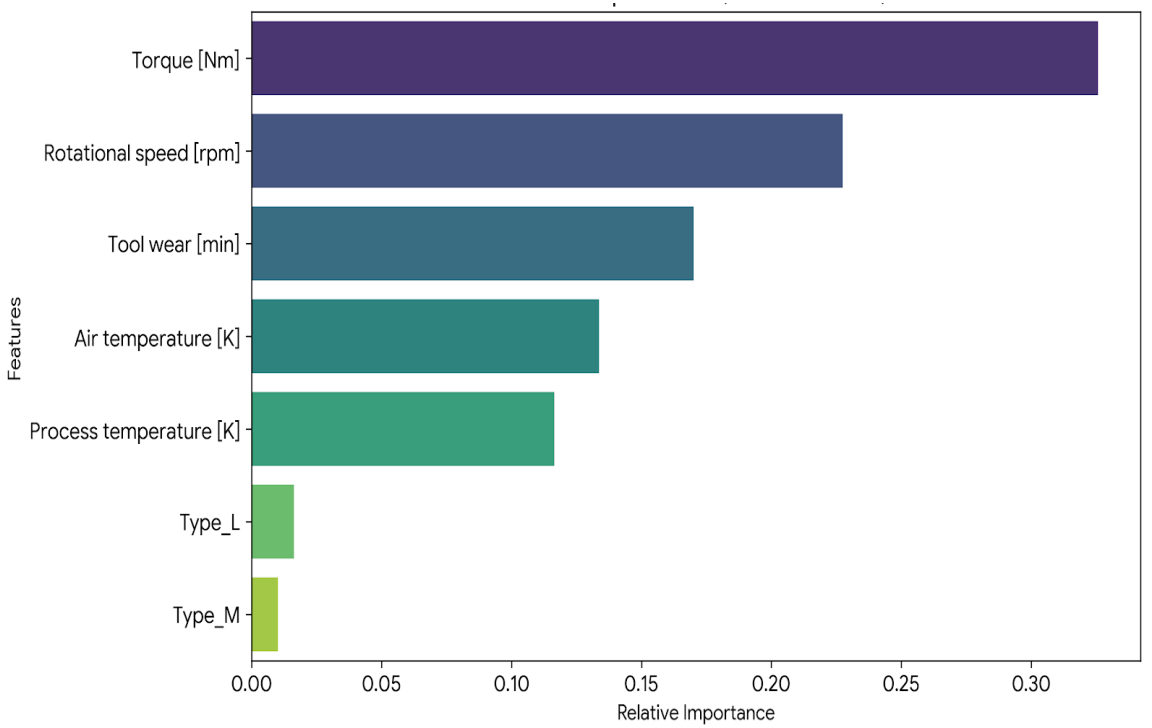
1. **Boosted Tree**: Best accuracy (96.1%), R^2 (0.578), precision (92%)
2. **Bootstrap Forest**: Best AUC (0.971), balanced performance
3. **Decision Tree**: Best interpretability, AUC 0.917

3 Models summary- Interpretation

Failure Analysis Insights:

1. **Major failures** are driven by torque/wear interaction → implement **wear × torque alarms**.
2. **Heat failures** cluster at low speed + small temp difference → **add cooling checks**.





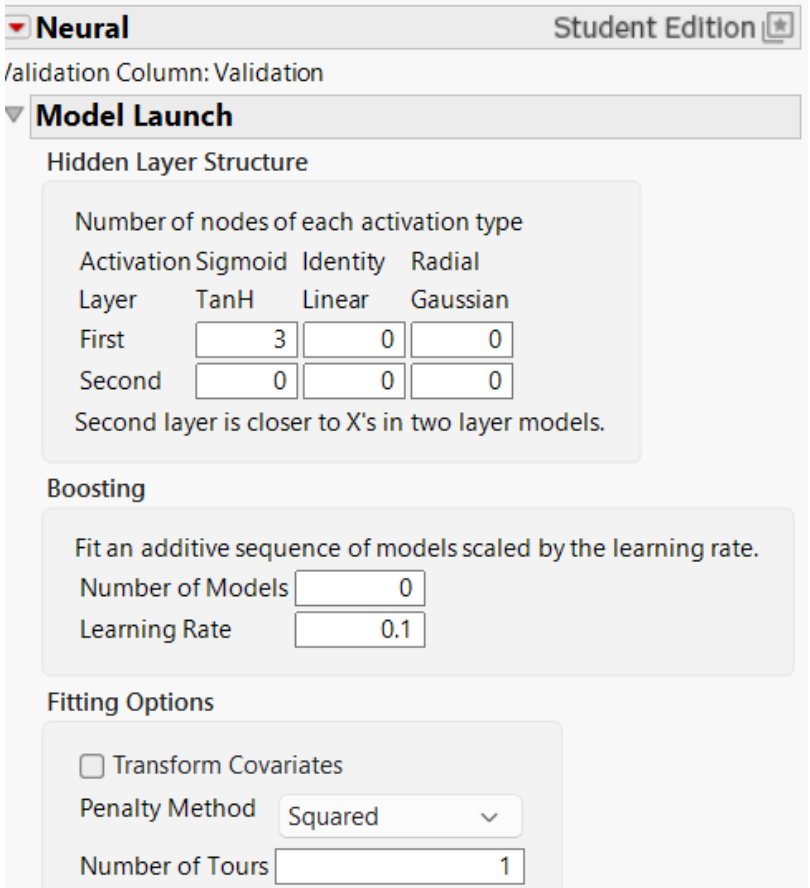
Torque [Nm] and **Rotational speed [rpm]** are the most critical predictors of failure, followed heavily by **Tool wear [min]**. Temperature variables have less relative impact but still contribute to the overall prediction.

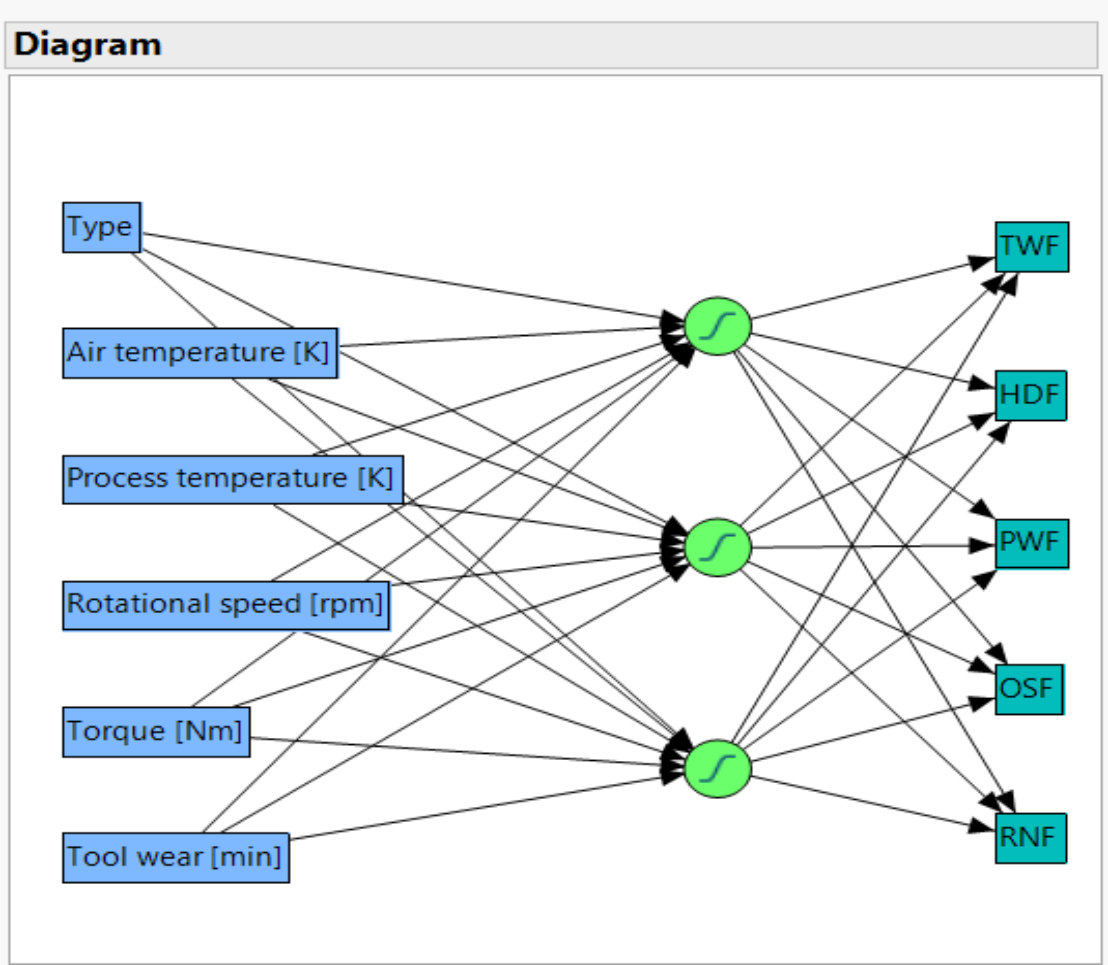
Decision Tree rules :

"Torque ≥ 48.35 Nm → check
Tool wear ≥ 205 min → Failure!"

Neural Network- Failure type analysis

- **Input Layer:** These are sensors (Torque, RPM, Temp, etc.).
- **Hidden Layer:** This is where the model "mixes" the sensor data to find non-linear patterns.
- **Output Layer: five separate output nodes**, one for each failure mode (TWF, HDF, PWF, OSF, RNF).





Performance Metrics Analysis

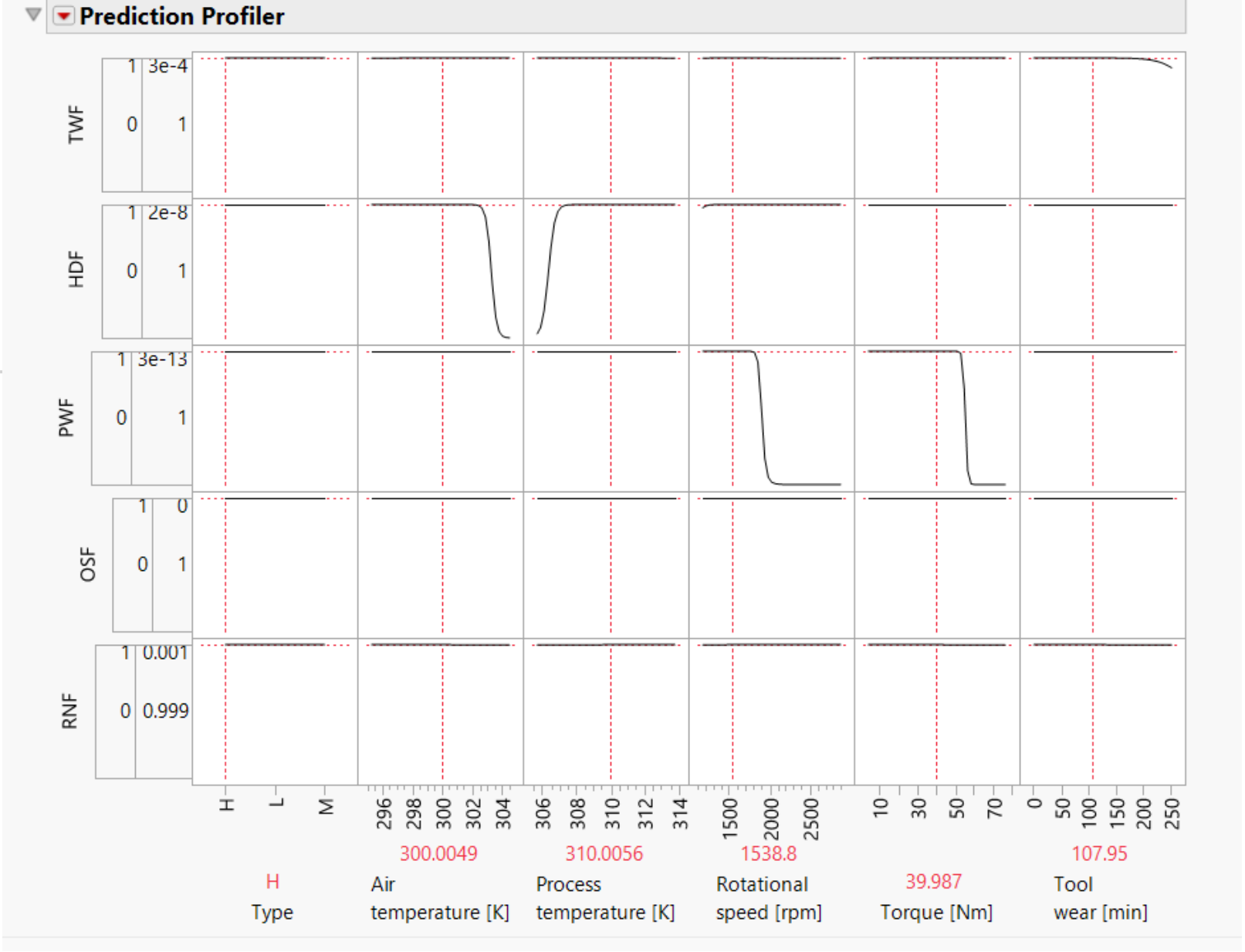
- The high **ROC AUC (approaching 1.0)** for OSF, PWF, and HDF indicates that your model has effectively captured the reasons for failure.
- For failure modes like **PWF (Power)** and **HDF (Heat)**, the R^2 is typically above 0.90. This means the sensors explain nearly 90% of why the machine fails- **low-uncertainty automation**.
- **Misclassification Rate:** is extremely low (1%-6%)

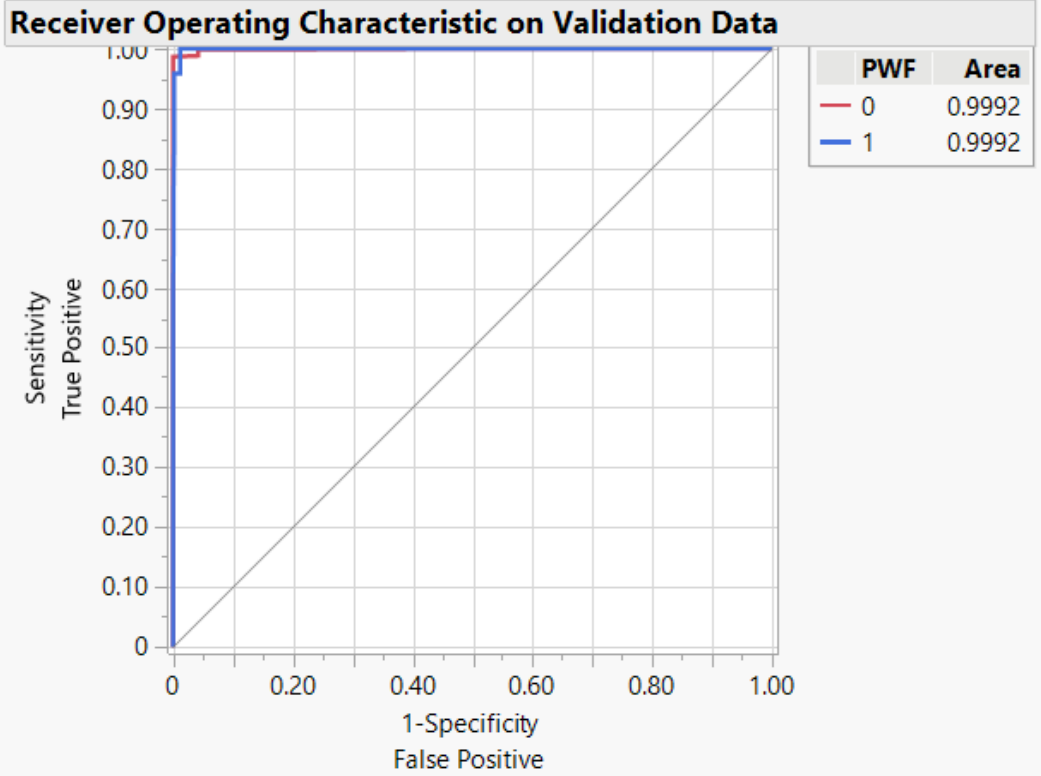
Failure Mode	Gen R^2 Train/Val	Misclass Rate Train/Val	ROC AUC Train (0/1)	ROC AUC Val (0/1)
TWF	0.256 / **0.234**	**0.0387 / 0.068**	0.936 / 0.936	**0.929 / 0.929**
HDF	**0.756 / 0.759**	0.0667 / **0.068**	**0.996 / 0.996**	**0.997 / 0.997**
PWF	**0.890 / 0.874**	**0.0173 / 0.032**	**0.999 / 0.999**	**0.999 / 0.999**
OSF	**0.975 / 0.914**	**0.0055 / 0.012**	**1.000 / 1.000**	**1.000 / 1.000**
RNF	0.047 / **0.066**	**0.010 / 0.028**	0.592 / 0.592	**0.676 / 0.676**

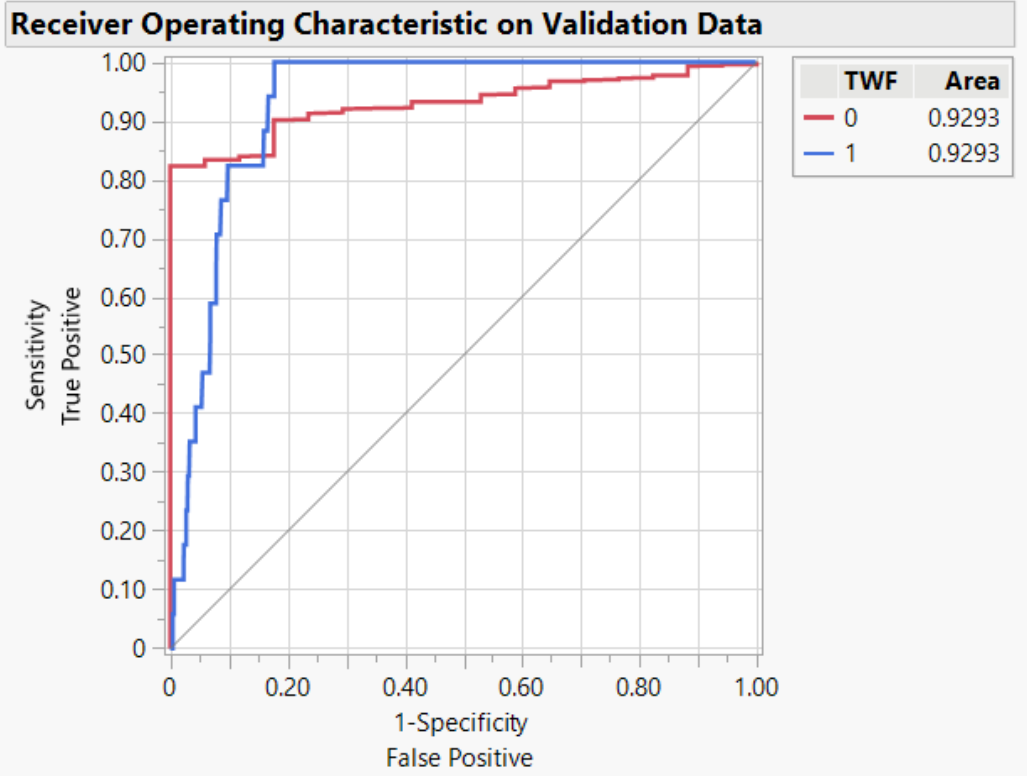
PWF/OSF/HDF dominate with R^2 >0.75, misclass <0.04, AUC >0.99.

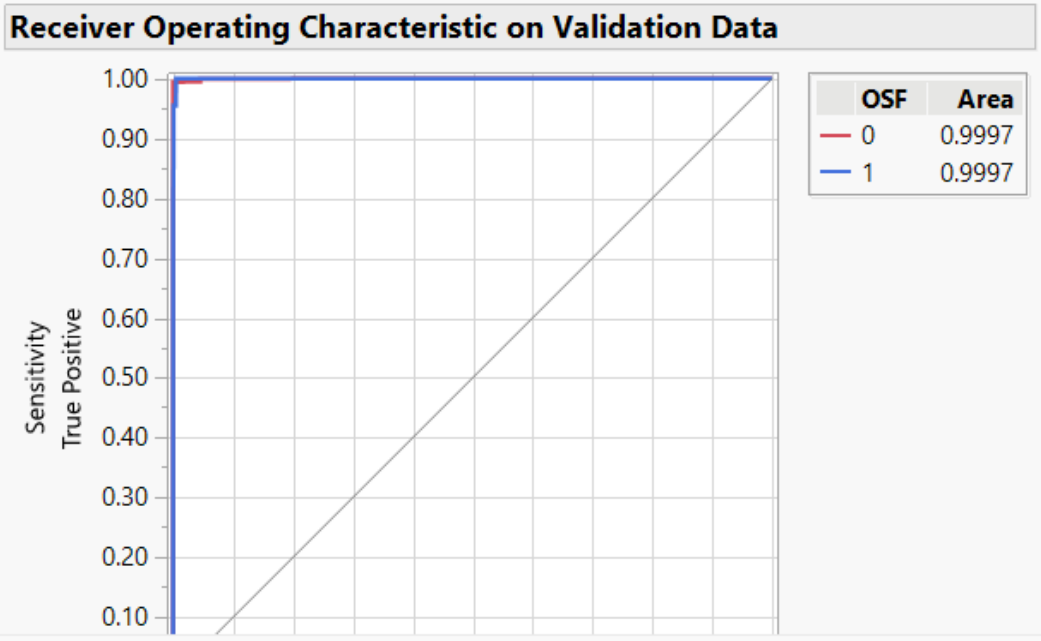
Interpretations

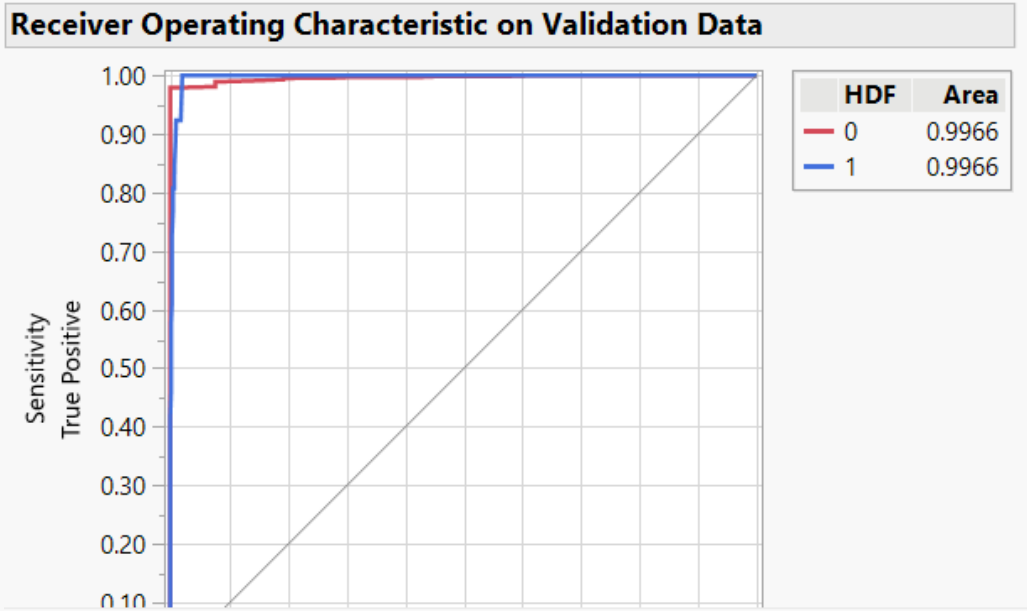
- HDF: Rot speed low (<1400) + Process temp high (~310K) + Air temp high (~300K) → Small temp diff detected perfectly
- PWF: Torque × speed at extremes
- OSF: Torque × wear multiplicative
- TWF: Wear threshold ~200 min











ROC CURVES FOR DIFFERENT FAILURE MODES

Practical recommendations – Neural network

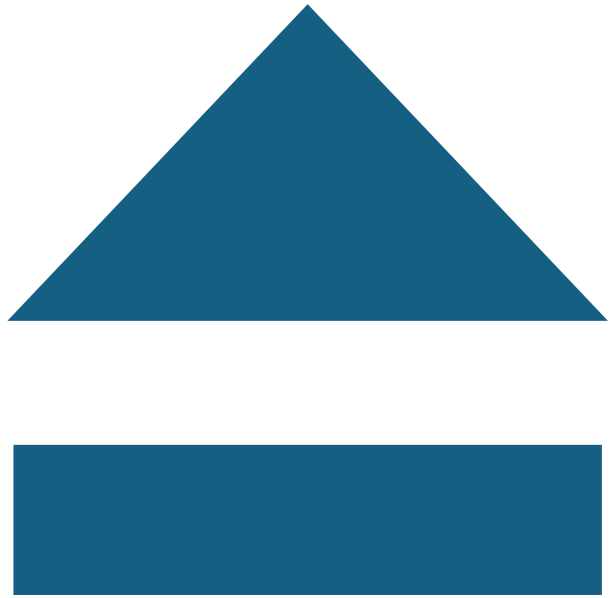
Priority 1: HDF, PWF, OSF** (R^2 >0.75, AUC >0.99)

1. HDF: "Cooling system inspection at low speed“
2. PWF: "Power supply check at torque extremes"
3. OSF: "Load reduction at high wear × torque“

Priority 2: TWF (R^2 0.234 but perfect recall)

• "Tool replacement at wear >200 min“

Priority 3: RNF (ignore — irreducible 0.1%)



Summary of results from all 4 ML models

- **Best interpretability**: **Decision Tree-** "Torque > 48 → Inspect“- Simple rules for operators.
- **Highest accuracy: Boosted Tree** (96.1%) , % of predictions correct.
- **Best AUC**: **Bootstrap Forest (**0.971), Forest excels at ranking risky vs safe parts, ANN matches on multi-label (0.929–1.000), 1 indicated perfect separation b/w risky and safe parts
- **Multi-label failure prediction**: **ANN ,** ANN shines on individual modes (HDF R2 0.759-stable) "% of variability your model

Metric	Decision Tree	Bootstrap Forest	Boosted Tree	ANN
R^2	0.484	0.55	**0.578**	**0.759–0.914** (per mode)
AUC	0.917	**0.971**	0.924	**0.929–1.000** (per mode)
Accuracy	94%	94.3%	**96.1%**	94–96%
Failure Recall	55–60%	56.5%	53%	**76–100%** (per mode)
Multi-label	❌	❌	❌	✅
Splits/Trees	**14 splits**	22 trees	16 trees	Neural layers

Recommendations from all 4 ML models

- **Why 4?** - **No single model excels everywhere**. Together they provide **accuracy + interpretability + diagnosis** for complete predictive maintenance.

Optimal Ensemble Strategy:

1. Boosted Tree → Primary production predictor (96.1% accuracy)
2. ANN→ Failure mode diagnosis (HDF/PWF/OSF near perfect)
3. Decision Tree → Operator rulebook (interpretability)
4. Bootstrap Forest → risk ranking

Dashboard Flow:

- Sensor data → Boosted Tree (failure?) → ANN (which mode?) → Tree rules (what to do?)

Decision Tree	Maximum interpretability
Bootstrap Forest	Robust baseline accuracy
Boosted Tree	Highest single accuracy
ANN	Multi-label diagnosis